	Class	Adm No
Candidate Name: ______________________________		

Promotional Examination II 2011
Pre-university 2

Biology Higher 1

Paper 1
Syllabus 8875/01

Friday | 23 September 2011 | 1 hour

Additional materials:
Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.
Write your name, Adm No. and class on all the papers you hand in.
Do not use staples, paper clips, highlighters, glue or correction fluid.

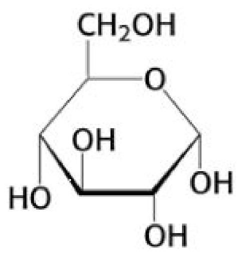
Paper 1
There are **thirty** questions in this paper. Answer **all** questions. For each question, there are four possible answers, **A, B, C** and **D**. Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Multiple Choice Answer Sheet.

Answer **all** questions in this paper.

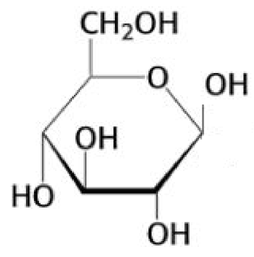
1. Which of the following best accounts for the difference in structure between amylose, which is helical, and cellulose which is fibrous?

   A. Presence or absence of 180^o rotation of alternate subunits
   B. Presence or absence of branching in the macromolecule
   C. Subunits are either monosaccharides or amino acids
   D. Different tendency of macromolecule to form hydrogen bonds with water.

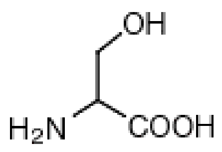
2. The figure below shows the structure of 4 monomers.



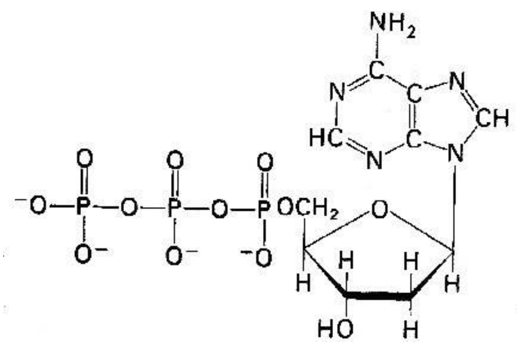
U



X



Y



Z

Which of the following combination of polymer, monomer and bond formed between monomers is correct?

	starch	cellulose	polypeptide	polynucleotide
A.	**X**: β-1,4 glycosidic bond	**U**: α-1,4 glycosidic bond	**Z**: ester linkage	**Y**: disulphide linkage
B.	**U**: α-1,4 glycosidic bond	**X**: β-1,4 glycosidic bond	**Y**: peptide bond	**Z**: phosphoester linkage
C.	**Y**: peptide bond	**X**: hydrogen bond	**Z**: ionic bond	**U**: hydrogen bond
D.	**X**: ionic bonds	**Y**: peptide bond	**U**: hydrogen bond	**Z**: α-1,6 glycosidic bond

3. Which combination describes a triglyceride?

	Soluble in water	Provides energy	Produces water when respired
A.	No	Yes	Yes
B.	Yes	No	Yes
C.	Yes	Yes	No
D.	No	Yes	No

4. Trypsin is a globular protein held together by several different types of bonds, giving the primary, secondary and tertiary levels of structure.

   Which combination correctly summarises the types of bond involved in each level of structure?

	Disulphide bonds	Hydrogen bonds	Ionic bonds	Peptide bonds
A.	Primary	Tertiary	Secondary	Tertiary
B.	Primary, tertiary	Primary	Secondary, tertiary	Secondary
C.	Secondary	Secondary	Tertiary	Primary
D.	Tertiary	Secondary, tertiary	Tertiary	Primary

5. The R group of the amino acid serine is –CH_2–OH. The R group of the amino acid alanine is –CH_3. Where would you expect to find these amino acids in a globular protein in aqueous solution?

   A. Serine would be in the interior, and alanine would be on the exterior of the globular Protein

   B. Alanine would be in the interior, and serine would be on the exterior of the globular protein.

   C. Both serine and alanine would be in the interior of the globular protein.

   D. Both serine and alanine would be in the interior and on the exterior of the globular protein.

6. A group of students investigated the digestion of lipids by lipase in the presence and absence of bile salts. Three test tubes were set up.

In tube **P**, milk was incubated with lipase only.
In tube **Q**, milk was incubated with lipase and bile salts.
In tube **R**, milk was incubated with bile salts only.

The results are shown as follows:

Time / minutes	**Mean pH**		
	Tube P	**Tube Q**	**Tube R**
0	8.5	8.5	8.5
10	8.0	7.7	8.5
20	7.6	7.0	8.5
30	7.3	6.5	8.5
40	7.0	6.5	8.5
50	6.5	6.5	8.5
60	6.5	6.5	8.5

Which of the following conclusions can be made?

I	The rate of lipid digestion is the highest in tube **Q**.
II	Bile salts act as coenzymes to increase the rate of lipid digestion.
III	The fall in pH in tubes **P** and **Q** is due to the rise in amount of fatty acids.
IV	The pH in tubes **P** and **Q** did not drop below 6.5 due to denaturation of lipase.

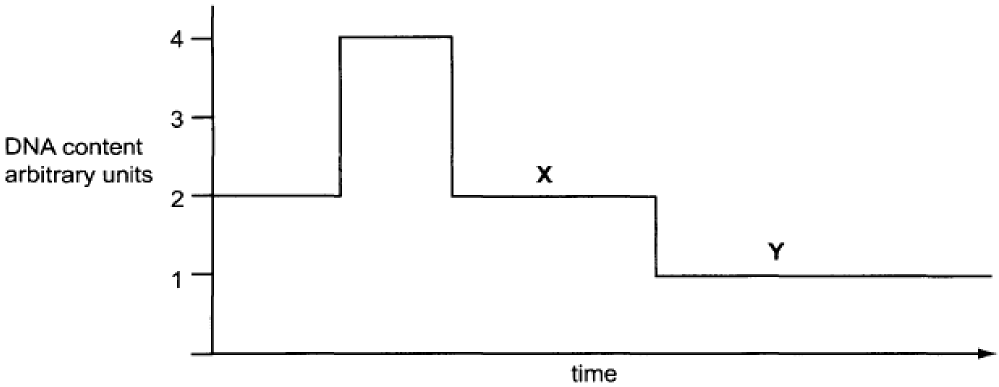
A. **I** and **II** only
B. **I** and **III** only
C. **II** and **III** only
D. **I**, **III** and **IV** only

7. The figure below shows a diploid plant cell undergoing anaphase.

Taxanes, which inhibit spindle fibres formation, are added to another cell from the same plant during anaphase II. If the amount of DNA in the diagram above is ***x***, which one of the following could be one possible result of a gamete formed?

	number of chromosomes	**amount of DNA**
A.	6	½ ***x***
B.	3	½ ***x***
C.	6	¼ ***x***
D.	3	¼ ***x***

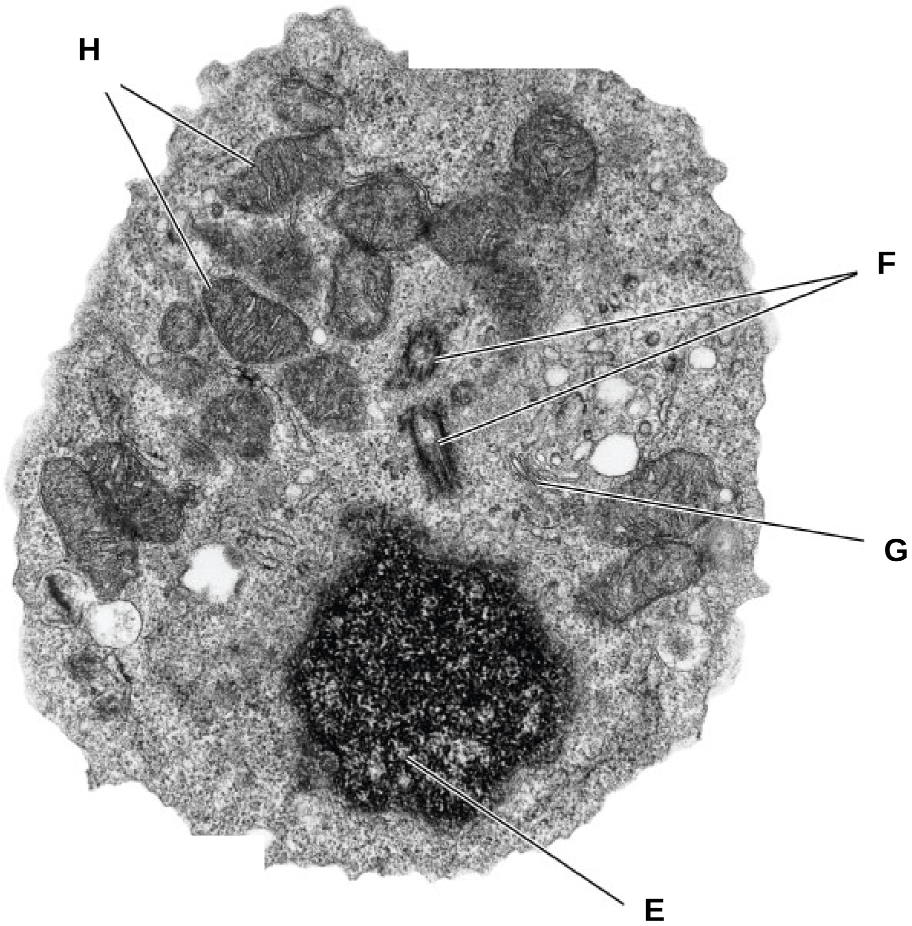
8. The graph shows the amount of DNA present in the nuclei of cells undergoing division in a multi-cellular organism.



Which combination is correct?

	Ploidy of cell at end of X	**Ploidy of cell at end of Y**
A.	diploid	diploid
B.	diploid	haploid
C.	haploid	haploid
D.	haploid	diploid

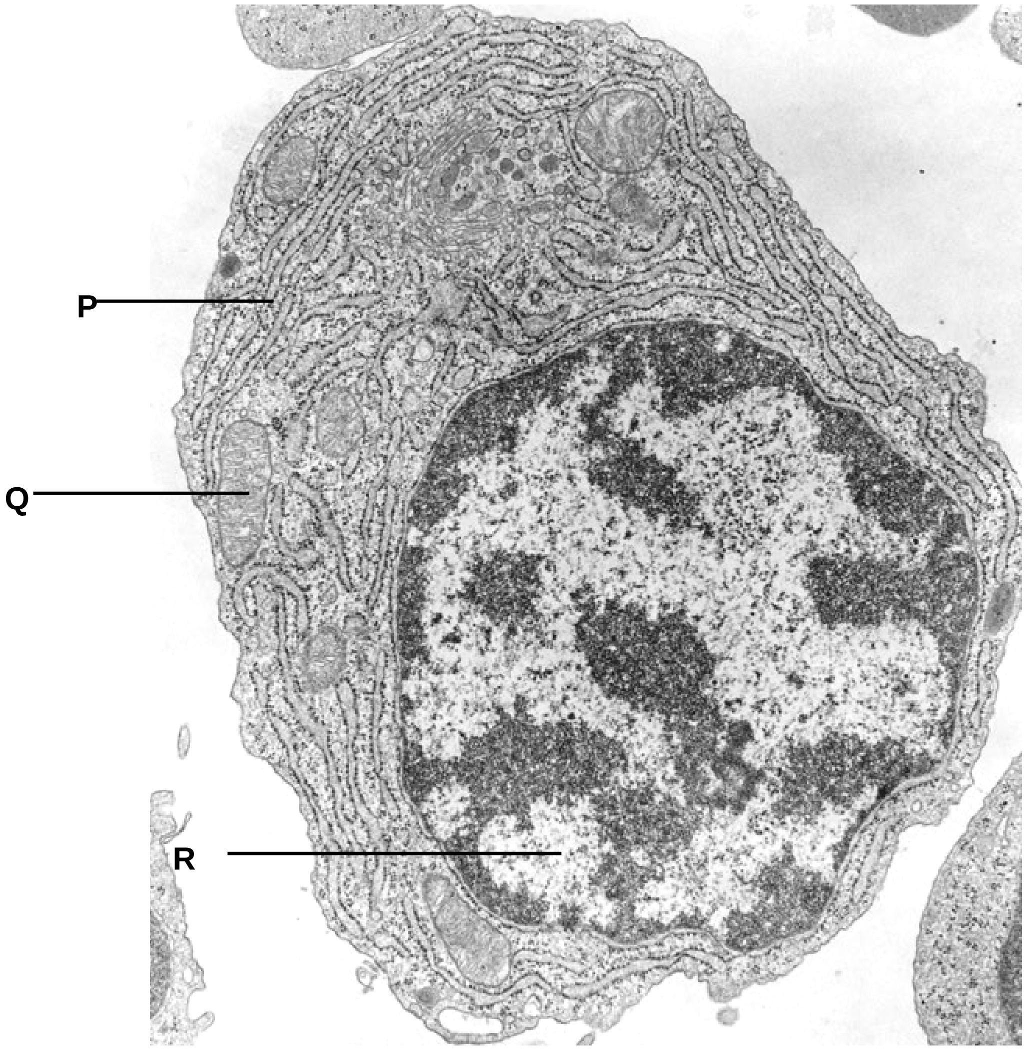
9. An electron micrograph of a cell is shown below.



Match the organelles **E, F, G and H** associated with the cellular processes listed.

	E	F	G	H
A.	DNA replication	Digestion of material	Organizes the spindle	Oxidative phosphorylation
B.	Oxidative phosphorylation	Organizes the spindle	Digestion of material	DNA replication
C.	Organizes the spindle	Digestion of material	Oxidative phosphorylation	Packaging of secretory products
D.	DNA replication	Organizes the spindle	Packaging of secretory products	Oxidative phosphorylation

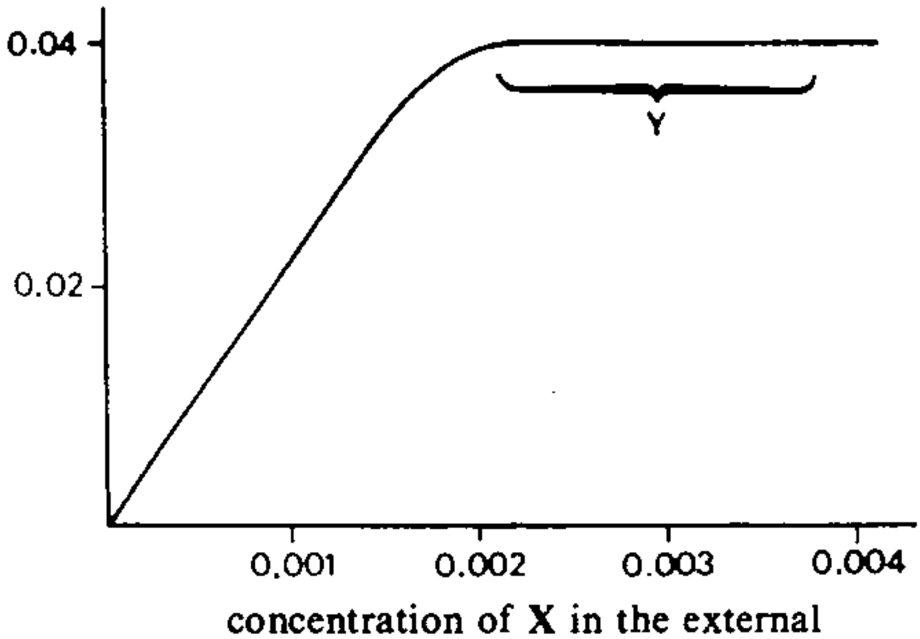
10. The figure shows an electron micrograph of a cell.



Which of the following about structures **P** – **R** are correct?

	P	Q	R
A.	Provides large surface area for attachment of ribosomes	Modification of transcripts	Active condensation of chromosome
B.	Transport of proteins to Golgi apparatus	Formation of ATP from light energy	Active Transcription of genes
C.	Synthesis and processing of membrane proteins	Sequestering of protons within a specialised compartment	Active Transcription of genes
D.	Prevent degradation by cytoplasmic enzymes	Oxidation of pyruvate	Active replication of DNA

11. Substance X (a mineral ion) is actively transported into cells. Equal-sized samples of cells were placed in media containing different concentrations of X for an hour. The intracellular concentration of X was then measured. All other metabolic conditions were maintained at the optimum level. The graph below shows the results.



From the information given above, which one of the following would account for the level region of the graph labelled Y?

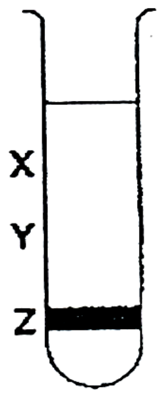
A. A respiratory inhibitor had been introduced.
B. All the active transport carriers had been operating at their maximum rate.
C. The active transport carriers had been inactivated by a non-competitive inhibitor.
D. As the internal concentration of X rose, more of the substance X was metabolised.

12. DNA polymerase III can only synthesize daughter DNA strand in the 5' to 3' direction. Which of the following is a consequence of this?

A. RNA primase needs to add an RNA primer first before DNA synthesis can begin
B. Replication error would not be proof-read by DNA polymerase III.
C. Complementary base pairing could then occur.
D. The lagging strand consists of short fragments of DNA.

13. A culture of bacteria had the entire DNA labelled with the heavy isotope of nitrogen, ^{15}N. The culture was then allowed to reproduce using nucleotides containing normal ^{14}N. The DNA was examined using a centrifuge after one generation and again after two generations.

The diagram shows the position of the DNA band at **Z** in the centrifuge tube when the DNA was first labelled.



In which pattern would the DNA be found after the first and after the second cell generations?

	After first generation	After second generation
A.	Half at **X** and half at **Y**	Quarter at **X** and at **Z** and half at **Y**
B.	Half at **X** and half at **Z**	Quarter at **X** and at **Z** and half at **Y**
C.	All at **Y**	Half at **X** and half at **Y**
D.	All at **X**	Half at **X** and half at **Y**

14. The diagram shows part of the normal sequence of an mRNA molecule.

CCAAGUGGUCCGCUAAAAUGGC

A mutation in the DNA resulted in a polypeptide beginning with the following sequence.

glycine – serine – proline – glycine – isoleucine

The DNA triplets for some amino acids are

Glycine	Isoleucine	Leucine	Proline	Serine
CGA	ATA	TTA	CCA	TCA
GGT	ATT	CTT	CCG	TCG
GGC		CTC		

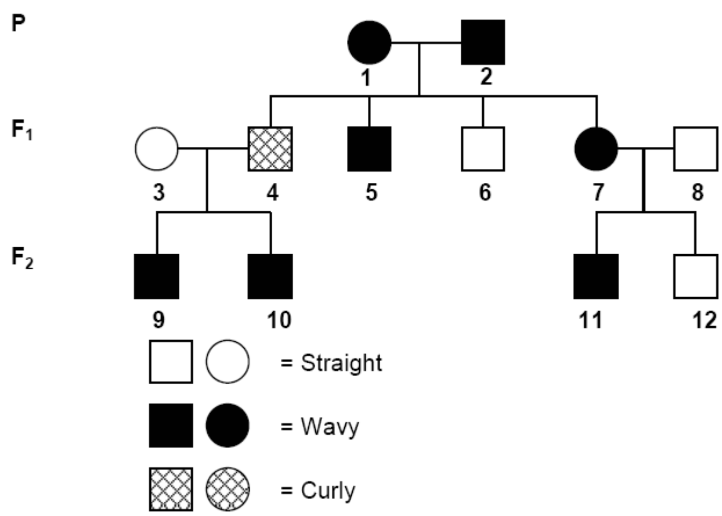
A. A reversal in the order of nucleotide

B. An addition of an extra nucleotide

C. The deletion of one nucleotide

D. A substitution of one nucleotide by a different nucleotide

15. Refer to the pedigree chart below, which of the following statements is true?



A. Wavy hair is dominant

B. Individual **1** is homozygous

C. Individual **4** is homozygous

D. Curly hair is recessive to straight hair

16. In fruit flies, body colour and eye colour are governed by different gene loci. A cross between yellow-bodied white- eyed females and grey-bodied red-eyed males produced females which were grey-bodied and red-eyed. When the F_1 females were backcrossed with their male parents, the resulting F_2 males comprised the following:

1330 yellow-bodied white-eyed flies,
26 yellow-bodied red-eyed flies,
35 grey-bodied white-eyed flies, and
1297 grey-bodied red-eyed flies

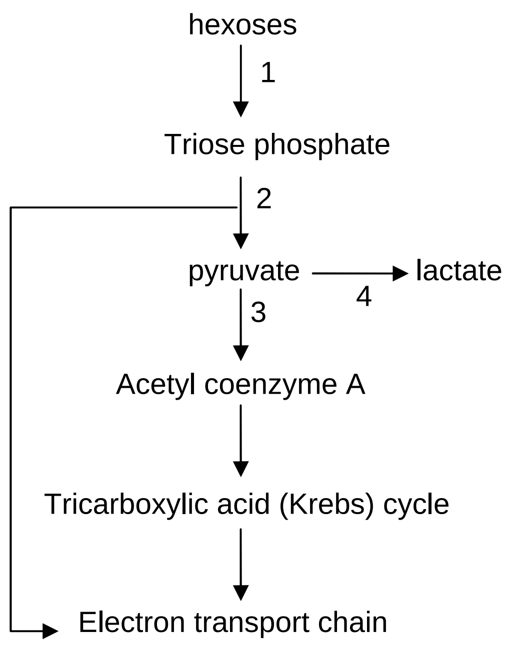
Which of the following explains the above results?

A. In fruit flies, the male is the homogametic sex
B. The two gene loci are sex-linked and crossing-over had taken place during gamete formation.
C. The two gene loci segregate by independent assortment
D. The allele for grey body is dominant over the allele for yellow body and the allele for white eyes is dominant over the allele for red eyes.

17. The genotypes of a tomato variety were **TtHh**, **Tthh**, **ttHh** and **tthh**, where **T** = tall and **H** = hairy. Which one of the following crosses would produce progeny giving a phenotypic ratio approximating to 1:1:1:1?

A. **TtHh** x **TtHh**
B. **TtHh** x **Tthh**
C. **TtHh** x **tthh**
D. **TTHH** x **tthh**

18. The diagram below summarises the stages involved in respiratory metabolism of mammalian skeletal muscle.



Stage 4 in the diagram occurs in mammalian muscle cells in the absence of oxygen. Why is stage 4 necessary?

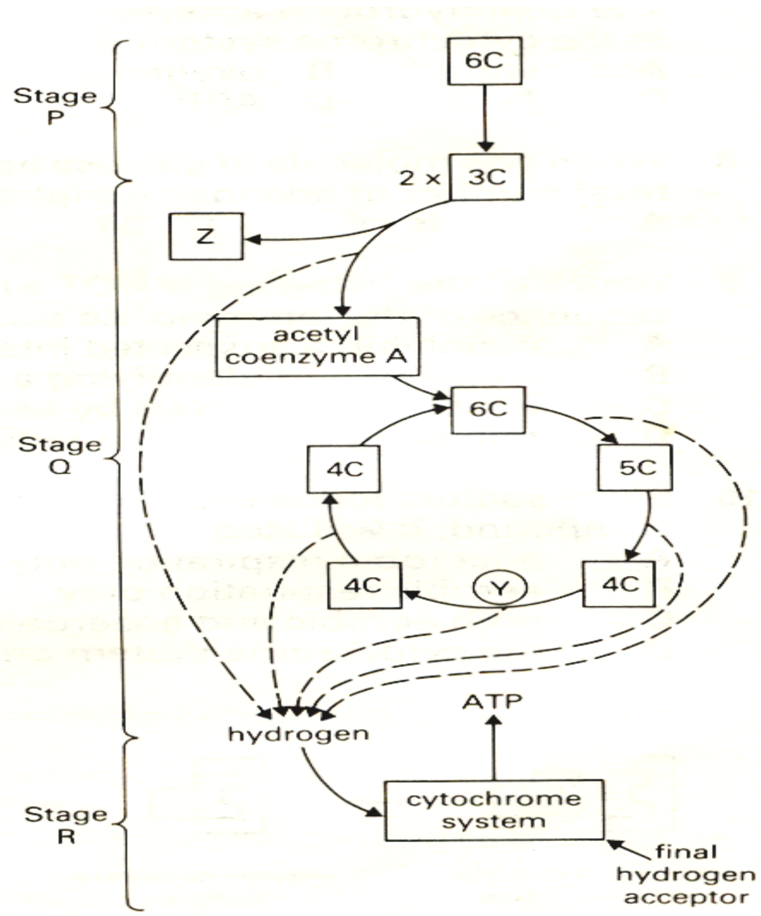
A. to oxidise the reduced nicotinamide adenine dinucleotide facilitating stage 2

B. to detoxify pyruvate

C. to form lactate which is immediately reconverted to glycogen

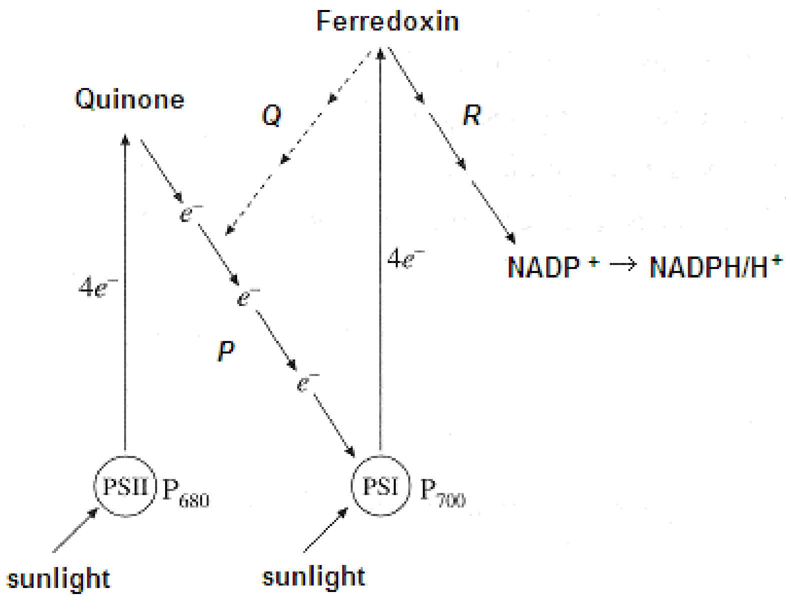
D. to provide reduced nicotinamide adenine dinucleotide for the electron transport chain

19. Which of the following indicates the correct locations at which stages P, Q and R occur in a cell?



	Cristae	Cytosol	Matrix
A.	R	Q	P
B.	Q	P	R
C.	P	R	Q
D.	R	P	Q

20. Figure shows the scheme for cyclic and non-cyclic phosphorylation.



Which of the following statements are true about the Z scheme?

I. $NADP^+$ is oxidized in non-cyclic phosphorylation.

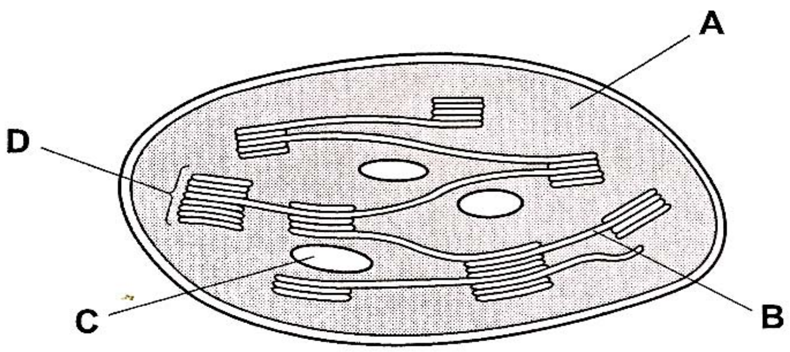
II. P_{680} and P_{700} are reduced after the electrons are excited to higher energy levels.

III. ATP is synthesised in step **P**.

IV. The products of non-cyclic phosphorylation are $NADPH/H^+$, ATP and oxygen.

A. **I** and **II**
B. **III** and **IV**
C. **I**, **II** and **III**
D. **II**, **III** and **IV**

21. The diagram shows a section through a chloroplast. Where the products of photophosphorylation would be used?

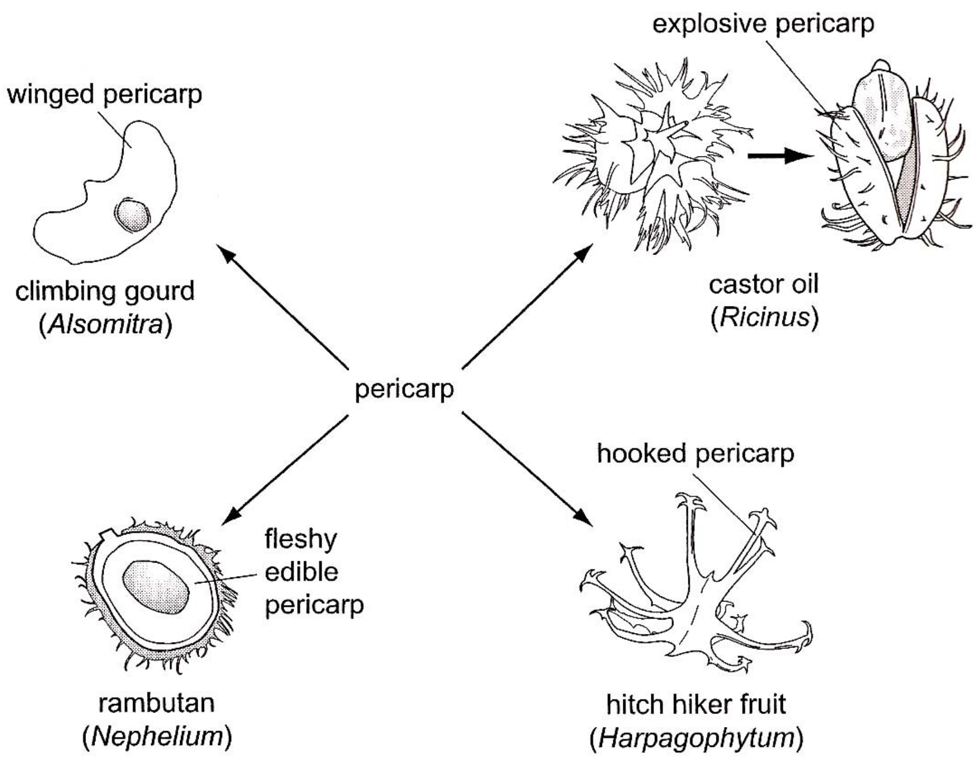


22. Given a population that contains genetic variation, what is the correct sequence of the following events, under the influence of natural selection?

1. Differential reproduction occurs.
2. A new selective pressure occurs.
3. Allele frequencies within the population changes.
4. Environmental change occurs.

A. **2**, **4**, **1**, **3**
B. **4**, **2**, **1**, **3**
C. **4**, **1**, **2**, **3**
D. **2**, **4**, **3**, **1**

23. The diagram illustrates variation in the pericarp (fruit wall) for a variety of methods of seed dispersal.



What do these examples illustrate?

A. The adaptive radiation of analogous structures showing convergent evolution
B. The adaptive radiation of analogous structures showing divergent evolution
C. The adaptive radiation of homologous structures showing convergent evolution
D. The adaptive radiation of homologous structures showing divergent evolution

24. Which of the following would provide the best information for distinguishing phylogenetic relationships between several species that are almost identical in anatomy?

A. The fossil records.
B. Homologous structures.
C. Comparative embryology.
D. Molecular comparisons of DNA and amino acid sequences.

25. At the start of the polymerase chain reaction (PCR), single stranded primers are added to the denatured DNA and the mixture cooled to 60^0C. What explains why the denatured DNA strands anneal with primers and **not** each other?

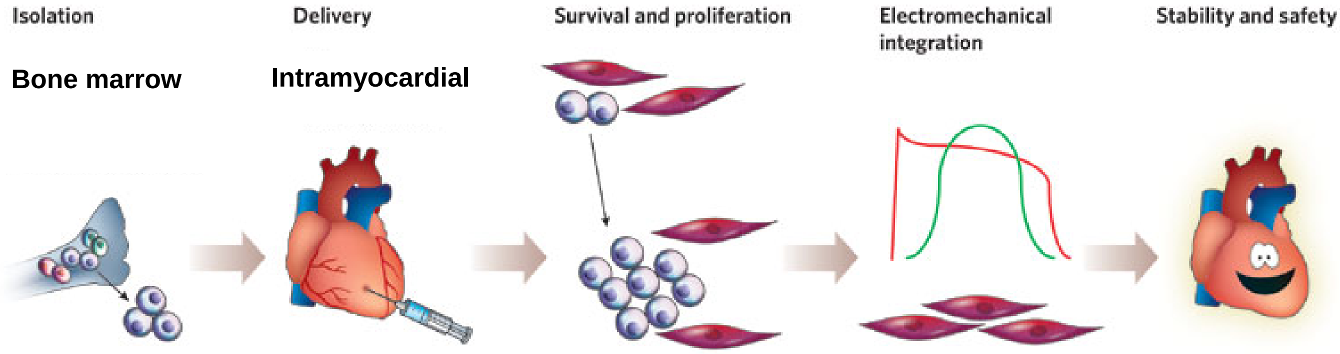
A.	The primers are shorter and anneal more easily
B.	The primers anneal only to the 3' end of the denatured DNA
C.	The primer concentration exceeds the denatured DNA concentration
D.	The temperature prevents the denatured DNA from annealing together

26. Which of the following shows the correct sequence of events for producing human growth hormone (hGH) using the bacterium *E. coli*?

I	Mix *E. coli* plasmids and human DNA together in the presence of DNA ligase.
II	Allow transformed *E. coli* to reproduce.
III	Treat *E. coli* plasmids and human DNA with the same restriction enzyme.
IV	Isolate *E. coli* plasmids and human DNA carrying the hGH gene.
V	Introduce recombinant plasmids into *E. coli.*

A. **V**, **IV**, **I**, **III**, **II**
B. **IV**, **I**, **III**, **V**, **II**
C. **IV**, **III**, **V**, **I**, **II**
D. **IV**, **III**, **I**, **V**, **II**

27. Stem cells hold promises for repairing damaged tissues. Recently the use of bone marrow blood stem cells has helped in cardiac repair as shown in the diagram below.



Which of the following conventional notion of stem cells is/are challenged as a result of this experiment?

i. The ability of stem cells to self renew by mitosis

ii. The role of the local tissue environment in stem cell differentiation.

iii. The effectiveness of adult stem cells.

iv. The multipotency of bone marrow blood stem cells.

A. iii only

B. iii and **iv**

C. iv only

D. All of the above

28. A student performed the following steps in a Southern blot experiment to determine the number of copies of a particular gene that has been inserted in a genetically modified organism.

i. Transfer of DNA to nitrocellulose membrane.

ii. Restriction digestion of genomic DNA.

iii. Cleaved DNA separated using gel electrophoresis.

iv. Create radioactive probe.

v. Incubate probe and membrane.

Which is the correct sequence to the above steps?

A. ii → iii → i → iv → v

B. ii → iii → i → v → iv

C. iv → v → i → ii → iii

D. v → iv → iii → ii → i

29. You are attempting to clone a gene of interest into the *lacZ* gene of a plasmid containing an ampicillin-resistance gene. The recombinant molecule was then introduced into a bacterial host sensitive (i.e. not resistant) to ampicillin. You grow your potential transformants on media containing both ampicillin and X-gal. Which of the following results would most likely indicate a successful clone?

A. cells which do not grow on the ampicillin plates
B. cells which form white colonies on the ampicillin plates
C. cells which form blue colonies on the ampicillin plates
D. cells which form light blue (half blue, half white) colonies on the ampicillin plates

30. Each of the statements is concerned with the success or failure of gene therapy as a permanent cure for genetic disorders.

Which statement(s) explain the limited success of gene therapy as a permanent cure for genetic disorders in human population?

i. Somatic cells cannot pass the modified gene on to any offspring
ii. The treatment does not last long as the treated cells die and are replaced
iii. The treatment does not result in all cells having the therapeutic gene

A. **i** only
B. **i** and **iii** only
C. **ii** and **iii** only
D. all of the above

End of Paper